

Multi-Phase Workflow: DnaA / Replication Initiation Model

Study 1: DnaA expression dynamics

Objective

Add dnaA transcription, mRNA dynamics, translation, and DnaA protein abundance dynamics sufficient to maintain realistic DnaA levels during steady growth.

Required Biological knowledge:

- dnaA promoter dynamics.
- DnaA autorepression at dnaAp
- Role of two consensus DnaA boxes and two DnaA-ATP boxes (maybe considered after Phase 2) in the dnaA promoter region
- Approximate cellular DnaA abundance from mass-spec or EcoCyc

Primary sources:

- replication_initiation_molecular_info.pdf

Expert questions:

- Ask Katayama and Lobner-Olesen labs for opinions on dnaA autorepression.
- Ask specifically about each DnaA boxes in the dnaA promoter region.

Implementation Tasks

1. Add dnaA transcription events.
2. Add dnaA mRNA degradation.
3. Add DnaA translation.
4. Add DnaA protein dilution/degradation if appropriate.
5. Add promoter occupancy state for dnaAp DnaA boxes.
6. Add placeholder autorepression:
 - if dnaAp occupancy increases, transcription rate decreases.
 - expose repression strength as a tunable parameter.

Read outs/ Visualization:

- transcription events vs time.
- mRNA number vs. time;
- mRNA concentration vs. time;
- DnaA number vs. time;
- DnaA concentration vs. time;
- Occupancy of DnaA boxes within dnaAp

Expected Behavior:

- Get the DnaA level within range (based on mass spec? Ecocyc number?). Expected: ~300-800 per cell (by mass spec, also from Ecocyc), or ~200 molecules per oriC (by western blots)
- roughly constant DnaA mRNA and protein concentrations (+/- 10%).
- If you stop synthesized DnaA, look for decrease in concentration
- In wildtype, after initiation look for DnaA change due to increased gene dosage and increased auto-repression strength.
- Move DnaA promoter region further from oriC to study gene dosage effect.

References:

- <https://www.pnas.org/doi/abs/10.1073/pnas.2322772121>
- Mass spec literatures, for protein abundance such as SeqA, DiaA, etc:
<https://www.nature.com/articles/nbt.3418>,
[https://www.cell.com/fulltext/S0092-8674\(14\)00232-3](https://www.cell.com/fulltext/S0092-8674(14)00232-3),
<https://link.springer.com/article/10.15252/msb.20209536>
- Check also Chenli Liu's Nature Microbiology for DnaA stoichiometry/mass-spec data:
<https://www.nature.com/articles/s41564-020-0717-x>
- Original DnaA protein abundance measurement:
<https://journals.asm.org/doi/abs/10.1128/jb.173.16.5194-5199.1991>
- DnaA protein stability literatures:
<https://journals.asm.org/doi/full/10.1128/jb.181.18.5557-5562.1999>,
<https://www.frontiersin.org/journals/molecular-biosciences/articles/10.3389/fmolb.2021.629953/full>.
- Key review paper:
<https://www.frontiersin.org/journals/microbiology/articles/10.3389/fmicb.2018.00319/full>.

Study 2: DnaA-ATP intrinsic hydrolysis activity

Required knowledge/mechanisms:

- Exponential decay $\text{DnaA-ATP} \rightarrow \text{DnaA-ADP}$ as simplest assumption
- DnaA binding to ATP and ADP strongly with similar affinities (ATP is moderately higher)
- If ratios are not within expected range it may be necessary to add a heuristic that holds them within an expected range, in order to move to the next phase

Primary sources:

- replication_initiation_molecular_info.pdf

Read outs/ Visualization:

- Transcription events per unit time (y) vs time (x).

- DnaA-ATP/DnaA-ADP/apo-DnaA concentration over time
- Background ATP, ADP readout

Expected behavior/ tests:

- Most DnaA is bound by either ATP and ADP
- If we have roughly constant DnaA concentrations over time, then expect DnaA-ATP constant over time. Expected range from our PNAS paper (Boesen et al. 2024):
DnaA-ATP/total DnaA = 0.2~0.5.
- Expect metabolism to have much higher ATP than ADP in the background.

References:

- Intrinsic activity:
<https://www.sciencedirect.com/science/article/pii/S0092867487902212?via%3Dihub>
- DnaA-ATP fraction indicated from Delta4 cells:
<https://www.pnas.org/doi/abs/10.1073/pnas.2322772121>.

Study 3: DnaA binding to DnaA boxes (oriC, chromosome, and dnaA promoter)

Required knowledge/mechanisms:

- All the dnaA boxes, and their affinities
- Assumptions at the first step: all 307 consensus DnaA boxes (TTWTNCACA) are high-affinity ($K_D \sim 1$ nM); oriC has 3-high + 8-low-affinity boxes ($K_D \sim 100$ nM); dnaA promoter has 2-high + 2-low-affinity boxes
- Assume all DNA binding does not alter DnaA-ATP hydrolysis rate
- Add DnaA binding according to affinity; both DnaA-ATP and DnaA-ADP can bind to high-affinity boxes – need to track these so we can release the correct forms later
- Low-affinity boxes ONLY bind to DnaA-ATP

Primary sources:

- replication_initiation_molecular_info.pdf

Read outs/ Visualization:

- Occupancy: oriC, high affinity boxes, low affinity boxes, total chromosome occupancy
- Free DnaA
- DnaA-ATP bound to DnaA boxes; DnaA-ADP bound to DnaA boxes; and to oriC and dnaA promoter
- A full chromosome visualization with DnaA boxes bound and unbound by DnaA-ATP or DnaA-ADP

Expected behavior/ tests:

- Unoccupied vs occupied sites on the chromosome over time – expect occupied to increase

- unoccupied/occupied oriCs over time – should remain unoccupied until most chromosome sites are occupied
- When unoccupied chromosome sites are ~10 (within order of magnitude of unoccupied oriC boxes), we should see oriC binding begin starting from the high-affinity sites.
- Free DnaA increases after titration, before initiation
- OriC occupancy: first low, and then high – should appear as a 2-step process
- Perturbation: start with high concentration of DnaA -> expected low activity of dnaA promoter. Look for autorepression behavior
- Compare oriC occupancy with mass initiation trigger to see if it roughly lines up (might not)

References:

- Modeling papers considering initiator titration:
<https://www.nature.com/articles/s41467-022-33886-6>,
<https://journals.aps.org/prxlife/abstract/10.1103/PRXLife.1.013011>,
<https://www.sciencedirect.com/science/article/pii/S0923250891900256>.

Study 4: Replace replication initiation heuristic with mechanism

Required knowledge/mechanisms:

- To start with, only consider slow growth with non-overlapping cell cycles
- Assume constant replication speed matching a 40mins C period (the period to replicate a whole chromosome); assume constant D period (the period between termination and cell division)
- Plot distribution of initiation mass over different random seeds. Initiation mass reaches a steady state regardless of initiation conditions
- As the replisome advances, the DnaA needs to be removed from the DnaA boxes because it depends on the double stranded DNA.

Primary sources:

- replication_initiation_molecular_info.pdf

Read outs/ Visualization:

- Replication event timing and mass; compare emergent timing and mass with heuristic initiation model.

Expected behavior/ tests:

- Stable cell cycle behavior (initiation once per generation on average)
- DnaA unbinds from chromosome domain when replisome fork moves passed it; DnaA re-binds to new dnaa boxes on new chromosome domains. Either DnaA-ATP or DnaA-ADP is removed, based on what was initially bound.
- Replication times are within a narrow distribution
- Timing difference between initiation events (less coordination at this point)

- Run in different growth conditions, and check when replication happens. Fast growth should lead to early replication
- Adjust number of dnaA boxes that can be bound to (set affinity to 0), measure time between initiation events.
 - increase/decrease number of DnaA boxes
 - Change distribution of DnaA boxes over chromosome
- 1/N scaling DnaA boxes vs timing between initiation events

References:

Study 5: Extrinsic DnaA-ATP/ADP conversion (RIDA, DDAH, DARS1&2)

Required knowledge/mechanisms:

- RIDA is replisome-dependent (activity proportional to replication fork number); DDAH activity depends on datA loci copy number; DARS1 and DARS2 activity depends their loci copy number, respectively.
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Read outs/ Visualization:

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Expected behavior/ tests:

- Potential increased synchrony of initiation timing.
- DnaA-ATP concentration oscillations
- DnaA-ATP vs replisomes

References:

Study 6: SeqA sequestration at oriC and dnaA promoter

Required knowledge/mechanisms:

- Cooperativity

Read outs/ Visualization:

Expected behavior/ tests:

References:

Phase 7: Cooperativity

Required knowledge/mechanisms:

Read outs/ Visualization:

Expected behavior/ tests:

References:

Phase 8: Validation

Required knowledge/mechanisms:

Read outs/ Visualization:

Expected behavior/ tests:

References: